

Handin 1: Introduction and Concepts

8/28/2026

Attempt 1



In Progress

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Details

Solve the following tasks and submit a PDF of the solutions. You can write the solution in English or in Norwegian.

You must have 50% correct answers to pass. Tasks marked with (*) are more difficult.

- (*) There are three people (Alex, Ben, and Cody), one of whom is a knight, one a knave, and one a spy. The knight always tells the truth, the knave always lies, and the spy can either lie or tell the truth. Alex says: "Cody is a knave." Ben says: "Alex is a knight." Cody says: "I am the spy." Who is the knight, who is the knave, and who is the spy? Explain your logical reasoning.
- Use truth tables to verify the following logical equivalence:

$$[p \rightarrow (q \vee r)] \Leftrightarrow [(p \wedge \neg q) \rightarrow r].$$
- (*) Give the reasons for the steps verifying the following proof of $s \vee t$. Follow the approach shown in the book in Example 2.24 (page 73) and use the rules of Table 2.19, page 78. You can also use the equivalences on pages 58, 59, and 62.

Our premises are (a) p , (b) $p \rightarrow q$, (c) $s \vee r$, (d) $r \rightarrow \neg q$. We use the following steps.

- p
- $p \rightarrow q$
- q
- $r \rightarrow \neg q$
- $q \rightarrow \neg r$
- $\neg r$
- $s \vee r$
- s
- $s \vee t$

- For the universe of all integers, let $p(x)$, $q(x)$, $r(x)$ be the following open statements.
 $p(x) : x > 0$, $q(x) : x \text{ is even}$, $r(x) : x \text{ is divisible by } 7$.

Write the following statements in symbolic form.

- At least one integer is even.
- There exists a positive even integer.
- No even integer is divisible by 7.
- All positive integers are even.

e. There exists an even integer divisible by 7.

Find all statements that are false and provide a counterexample for each of those.

5. Using only the Peano definitions of addition and multiplication (see below), prove that for all natural numbers n it is true that $1 \cdot n = n$.

Peano definitions: $0+a=a$ [+.1], $s(x)+a=s(x+a)$ [+.2], $0 \cdot a=0$ [.1], $s(x) \cdot a=(x \cdot a)+a$ [.2], $1=s(0)$ [n.1]

6. (*) Find the flaws in the following answer and proof, and fix them.

Question: For which natural numbers n is $n + n = n \cdot n$?

Answer: $n = 2$

Proof:

$n + n = n \cdot n$ given

$n \cdot 2 = n \cdot n$ definition of multiplication

$2 \cdot n = n \cdot n$ commutativity of $\cdot$

$2 = n$ division by n

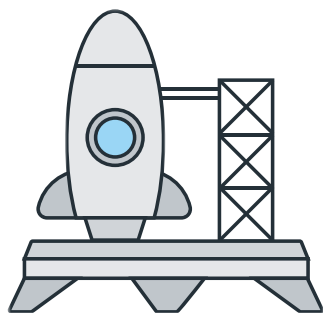
$n = 2$ commutativity of $=$

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